RESIDUE FIELD EXTENSIONS OF A GALOIS EXTENSION OF DVRS ARE GALOIS WITH SURJECTIVE DECOMPOSITION MAP

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Allowed standard prerequisites (editorial).

- The decomposition group acts with one orbit on the primes and induces all automorphisms of the residue field extension when the extension of the base valuation ring is considered. invoked at proof lines 33244-33245. Gives normality and surjectivity in parts (1) and (2).
- If one residue extension kappa(m)/kappa is separable then all are. invoked at proof lines 33245-33246. Justifies the equivalently-all clause and the Galois upgrade.

Source reference. Original lemma 0BRJ: sources/stacks-more-algebra/pinned-original-source.tex, source lines 32704–32704. Cited at proof lines 33244-33245 for parts (1) and (2); the conclusion is used inside the counted span.

Source reference. Original lemma 09EA: sources/stacks-more-algebra/pinned-original-source.tex, source lines 33179–33179. Cited at proof lines 33245-33246 for the equivalently-all clause; the conclusion is used inside the counted span.

Lemma 0.2. *Let A be a discrete valuation ring with fraction field K and residue field κ. Let L/K be a finite Galois extension with Galois group G. Let B be the integral closure of A in L. Let $\mathfrak{m}$ be a maximal ideal of B. Then*

(1) *the field extension $\kappa(\mathfrak{m})/\kappa$ is normal, and*
(2) *$D \to Aut(\kappa(\mathfrak{m})/\kappa)$ is surjective.*

If for some (equivalently all) maximal ideal(s) $\mathfrak{m} \subset B$ the field extension $\kappa(\mathfrak{m})/\kappa$ is separable, then

(3) *$\kappa(\mathfrak{m})/\kappa$ is Galois, and*
(4) *$D \to Gal(\kappa(\mathfrak{m})/\kappa)$ is surjective.*

Here $D \subset G$ is the decomposition group of $\mathfrak{m}$.

Proof. Observe that $A = B^G$ as A is integrally closed in K and $K = L^G$. Thus parts (1) and (2) follow from Lemma 0BRJ. The "equivalently all" part of the lemma follows from Lemma 09EA. Assume $\kappa(\mathfrak{m})/\kappa$ is separable. Then parts (3) and (4) follow immediately from (1) and (2). □

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